FOODBRIDGE — TECHNICAL ARCHITECTURE QUICK REFERENCE

Full-Stack Community Surplus Food Rescue & Algorithmic Expiry Platform • Release 1.0.0

1. PRODUCTION ARCHITECTURE TOPOLOGY

Tier	Host / Service	Runtime & Stack	Live URL / Coordinates
Frontend SPA	Vercel Global Edge	React 18.3, TypeScript 5.7, Vite 6.4	Deploying to Vercel (dist)
Backend REST API	Render Cloud Container	Node.js 24.22 LTS, Express 4.21, Mongoose 8.9	https://foodbridge-xnpd.onrender.com
Database Tier	MongoDB Atlas (AWS)	Replica Set M0 (Cluster0 / foodbridge)	Encrypted SRV (TLS Port 27017)
Source Repository	GitHub Cloud	Branch 'main', Git 2.45+	https://github.com/vishnu9392/FoodBridge

2. MONGOOSE DATABASE SCHEMA REFERENCE

Model	Key Attributes & BSON Types	Validation & Referential Integrity
User (users)	email (Str), passwordHash (Str), role (Enum: donor/recipient/admin), name (Str), organization (Str), phone (Str), address (Str), bio (Str)	Unique email index, bcrypt 10 salt rounds, role-based guard compatibility
Donation (donations)	donor (Ref User), foodName (Str), category (Enum 7), quantity (Num), availableQuantity (Num), reservedQuantity (Num), unit (Enum 5), prepDate (Date), expiryDate (Date), location (Subdoc: lat/lng/address), status (Enum 5)	Invariant: available + reserved <= total. expiryDate must be > prepDate. Filtered by active unexpired.
Pickup (pickups)	donation (Ref Donation), recipient (Ref User), donor (Ref User), requestedQuantity (Num), status (Enum 5), preferredDate/Time (Str), notes (Str)	State Machine: pending -> approved/rejected -> completed/cancelled. Deducts on approve, restores on cancel.

3. REST API ENDPOINT SPECIFICATION MATRIX (18 Endpoints)

Method & URL	Auth Guard	Role	Operation & Controller Logic
POST /api/auth/register	Public	Any	Hashes password with bcrypt, generates JWT session token (201 Created)
POST /api/auth/login	Public	Any	Verifies credentials, signs 7-day HMAC-SHA256 JWT token (200 OK)
GET /api/auth/me	Bearer	Any	Returns authenticated user profile from req.user decoded claims
PUT /api/auth/profile	Bearer	Any	Updates telephone, organization name, address, and mission bio
PUT /api/auth/password	Bearer	Any	Validates old password before hashing new credentials
GET /api/donations	Public	Any	Queries available, unexpired surplus items with category/search filters
GET /api/donations/:id	Public	Any	Returns comprehensive single donation details with coordinates and donor profile
GET /api/donations/user/my	Bearer	Donor	Queries all surplus listings published by logged-in donor
POST /api/donations	Bearer	Donor	Publishes surplus batch, validates timestamps and GPS location (201)
PUT /api/donations/:id	Bearer	Owner	Updates active donation record (guarded by donor ownership check)
DELETE /api/donations/:id	Bearer	Owner	Soft-cancels and removes active listing from marketplace
GET /api/pickups	Bearer	Any	Lists all pickup requests involving user as donor or recipient
POST /api/pickups	Bearer	Any	Submits pickup request, verifies requestedQuantity <= availableQuantity
PATCH /api/pickups/:id/status	Bearer	Involved	Drives state machine (approved, rejected, completed, cancelled)
GET /api/dashboard/stats	Bearer	Any	Aggregates rescue metrics (kg rescued, meals served, pending requests)
POST /api/expiry/calculate	Public	Any	Calculates delta milliseconds, status (fresh/expiring_soon/expired)
POST /api/expiry/estimate	Public	Any	Calculates shelf-life projection using temperature storage multipliers
GET /api/health	Public	Any	Diagnostics reporting uptime, Node version, and MongoDB connection state

4. STATE MACHINE INVARIANTS & VERIFICATION SUMMARY

State Machine Lifecycle: [pending] ➔ (Donor Approves) ➔ [approved] ➔ (Handover Verified) ➔ [completed].
Alternative Paths: [pending] ➔ (Donor Rejects) ➔ [rejected] | [pending/approved] ➔ (Party Cancels) ➔ [cancelled].
Inventory Guard: Approving a request atomically deducts requested portions from availableQuantity and adds them to reservedQuantity. Rejecting or cancelling restores portions back to availableQuantity.
Automated Test Suite: 20/20 unit tests passed (0 failures, 477 ms). Verified: past date expiry, 24h urgency threshold, unparseable dates, storage multipliers, donor approval permissions, terminal state immutability, email RFC regex, and positive quantity rules.